Representation of $\mathfrak{su}(2)$ by complexification

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1 Introduction

The Lie algebra $\mathfrak{su}(2)$ is associated with the special unitary group $SU(2)$, which plays a crucial role in various areas of mathematics and physics. Though contains all entries of complex number, it's actually a real Lie algebra. In fact, let $A = A_1 + iA_2 \in \mathfrak{su}(n)$, where i is the imaginary unit and A_1 and A_2 are both real matrix, by definition, we have

$$A + A^\dagger = (A_1 + A_1^T) + i(A_2 - A_2^T) = 0$$

which shows that A_1 is anti-symmetric and A_2 is symmetric, and that $tr(A) = tr(A_1) + i \cdot tr(A_2) = i \cdot tr(A_2) = 0$. Furthermore, $iA = -A_2 + iA_1 \in \mathfrak{su}(2)$ iff both A_1 and A_2 are symmetric and anti-symmetric simultaneously, namely, $A_1 = A_2 = A = 0$, which implies that $\mathfrak{su}(2)$ is not $\mathbb{C}$-linear.

Consider the case that $n = 2$, it's not hard to obtain the generators of $\mathfrak{su}(2)$, denoted by

$$\sigma_1 = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \sigma_2 = \frac{1}{2}\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \sigma_3 = \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

An easy computation tells us the commutation between these generators

$$[\sigma_i, \sigma_j] = \sigma_k$$

where (i, j, k) is the rotation of $(1, 2, 3)$, this relation gives an isomorphism to $\mathfrak{so}(3) \cong (\mathbb{R}^3, \times)$ by $\sigma_i \mapsto e_i$, where e_i is the standard basis of $\mathbb{R}^3$ with cross product .The structure and properties of $\mathfrak{su}(2)$ make it a vital object of study, linking abstract mathematical theory with practical physical applications. Its representation theory, commutation relations, and role in symmetry operations provide deep insights into the underlying symmetries of the universe.

In the following discussion, we will dig into the representation of $\mathfrak{su}(2)$ via complexification, which is right the $\mathfrak{sl}(2, \mathbb{C})$ and find out some interesting properties and conclusion.

2 Complexification Of Real Vector Space

2.1 Definitions of complexification

Let V be a finite-dimensional real vector space, we then use similar method to that we used to create complex number from real number and make it into a complex vector space, called complexification. In fact, we can view such complexification as a $\mathbb{C}$-module of the direct sum space $V \oplus V$, and draw a rigorous defination.

Definition 2.1. The complexification of a real vector space V, denoted by $V_\mathbb{C}$, is a $\mathbb{C}$-module of $V \oplus V$, s.t for arbitrary $v_1, v_2 \in V$ and $a + bi \in \mathbb{C}$

$$(a + bi).(v_1, v_2) = (av_1 - bv_2, av_2 + bv_1)$$

It's not hard to verify that such construction do make a $\mathbb{C}$-module, and we can simply denote each element by $v_1 + iv_2$, with the usual computation we did for complex number. In particular, we have

$$i(v_1 + iv_2) = i.(v_1, v_2) = (-v_2, v_1) = -v_2 + iv_1$$

$$a(v_1 + iv_2) = a.(v_1, v_2) = (av_1, av_2) = av_1 + i(av_2)$$

Let the first entry be zero, we can obtain that

$$a(iv) = a.(0, v) = (0, av) = i(av)$$

This shows that the dimension is invariant under complexification, namely, $\dim_{\mathbb{R}} V = \dim_{\mathbb{C}} V_{\mathbb{C}}$ In fact, if $\{v_i\}$ is a group of $\mathbb{R}$ basis for V, then it also makes up a group of $\mathbb{C}$ basis for $V_{\mathbb{C}}$.

Further consider V to be a real algebra, equipped with a $\mathbb{R}$-bilinear form m to itself , namely

$$m : V \times V \to V$$

We want to seek for a natural extension $m_{\mathbb{C}}$ for m to be the unique $\mathbb{C}$-bilinear form on $V_{\mathbb{C}}$ s.t the restriction to $V_{\mathbb{R}} := V \oplus \{0\} \cong V$ is m with real coefficient.

Proposition 2.2. *The extension $m_{\mathbb{C}}$ of m defined above exists uniquely.*

Proof. We claim that

$$m_{\mathbb{C}}(v_1 + iv_2, w_1 + iw_2) := (m(v_1, w_1) - m(v_2, w_2)) + i(m(v_1, w_2) + m(v_2, w_1))$$

is an admissible extension by analogy to operations of complex numbers. In fact, it's obviously $\mathbb{R}$-linear for the first position with the equation $a(iv) = i(av)$ above. On the other hand,

$$\begin{aligned}
m_{\mathbb{C}}(i(v_1 + iv_2), w_1 + iw_2) &= m_{\mathbb{C}}(-v_2 + iv_1, w_1 + iw_2) \\
&= (-m(v_2, w_1) - m(v_1, w_2)) \\
&\quad + i(m(v_1, w_1) - m(v_2, w_2)) \\
&= im_{\mathbb{C}}(v_1 + iv_2, w_1 + iw_2)
\end{aligned}$$

this proofs the $\mathbb{C}$-linearity for the first position and the same arguement goes for the second position, which validates the existence.

The uniqueness follows by the identity $(0, v) = i(v, 0) = iv$. To be more specificly, for each possible extension $m_{\mathbb{C}}$, with the $\mathbb{C}$-linearity we have

$$\begin{aligned}
m_{\mathbb{C}}(v_1 + iv_2, w_1 + iw_2) &= m_{\mathbb{C}}(v_1, w_1) - m_{\mathbb{C}}(v_2, w_2) \\
&\quad + i(m_{\mathbb{C}}(v_1, w_2) + m_{\mathbb{C}}(v_2, w_1)) \\
&= (m(v_1, w_1) - m(v_2, w_2)) \\
&\quad + i(m(v_1, w_2) + m(v_2, w_1))
\end{aligned}$$

which completes the proof.

□

We now generalize such idea of extension to arbitrary $\mathbb{R}$ -linear homomorphism from V to a complex vector space $f : V \to W$, in fact, the theorem below holds:

Theorem 2.3. *Let ι be the embbeding of V into its complexification $V_{\mathbb{C}}$ and f be as described above, then there uniquely exists a $\mathbb{C}$-linear mapping $\tilde{f} : V_{\mathbb{C}} \to W$ s.t the following diagram commutes:*

$$\begin{tikzcd} & V \arrow[dl, "\iota"'] \arrow[dr, "f"] & \\ V_{\mathbb{C}} \arrow[rr, "\tilde{f}"] & & W \end{tikzcd}$$

The proof is provided by notes of KEITH CONRAD from University of Connecticut. Consider a special case that the homomorphism here is the composition of a homomorphism to another real vector space and its embbeding to its complexification, that is, we have the diagram below commutes:

$$\begin{tikzcd} V \arrow[r, "f"] \arrow[d, "\iota_V"'] & W \arrow[d, "\iota_W"] \\ V_{\mathbb{C}} \arrow[r, "\tilde{f}"] & W_{\mathbb{C}} \end{tikzcd}$$

Apply this theorem to $\mathfrak{su}(2)$, we transform the study of the real Lie algebra $\mathfrak{su}(2)$ into its complexification, which turns out to be $\mathfrak{sl}(2, \mathbb{C})$.

2.2 The conjugation of complexification

As is known that there are three basic operation on comlpex number: taking the real part, the imaginary part and the conjugation. A natural idea is to generalize these notions to the complexification of a real vector space by abstract properties.

Definition 2.4. The conjugate operator on a complexification space $V_{\mathbb{C}}$ of V is a mapping, denoted by 'Con', that satisfies the following conditions:
- $\mathrm{Con}(v_1 + v_2) = \mathrm{Con}(v_1) + \mathrm{Con}(v_2)$, for $\forall v_1, v_2 \in V_{\mathbb{C}}$
- $\mathrm{Con}(zv) = \overline{z}\mathrm{Con}(v)$, for $\forall z \in \mathbb{C}$ and $v \in V_{\mathbb{C}}$
- $\mathrm{Con}^2 = \mathrm{Id}$
- $\mathrm{Con}|_V = \mathrm{Id}$

The first and second propertie together is called conjugate linearity. It' not hard to see that such mapping exists uniquely as long as one choose a basis of V, and that $v \in V_{\mathbb{C}} \Leftrightarrow \mathrm{Con}(v) = v$.

We now denote the conjugation of v by $\overline{v}$ following the convention in complex number. Another idea arise from here is that can we generalize conjugation to $\mathbf{Hom}(V_{\mathbb{C}}, W_{\mathbb{C}})$, here we give the defination by analogy to the equation $\overline{z_1 z_2} = \overline{z_1}\,\overline{z_2}$ for complex number.

Definition 2.5. The conjugation for $A \in \mathbf{Hom}(V_{\mathbb{C}}, W_{\mathbb{C}})$, denoted by $\overline{A}$, is the linearity mapping satisfying the equation

$$\overline{Av} = \overline{A}\,\overline{v}$$

for any $v \in V_{\mathbb{C}}$.

In fact, the defination is equivalent to the eqution:

$$\overline{A} = \mathrm{Con} \circ A \circ \mathrm{Con}^{-1}$$

Notice that Con is not $\mathbb{C}$-linear, however, $\overline{A}$ is. If we apply the eqution to scalar multiplication for $z \in \mathbb{C}$, we actually get its conjugation $\overline{z}$; Another useful properties derived from this eqution is that

Property 2.6.

$$\begin{aligned}\overline{\overline{A}} &= A \\ \overline{A+B} &= \overline{A}+\overline{B} \\ \overline{AB} &= \overline{A}\,\overline{B} \\ A \in \mathbf{Hom}(V,W) &\iff \overline{A} = A\end{aligned}$$

which coincide with the conjugation for $V_{\mathbb{C}}$. The fourth property comes from that for each $v \in V$, $\overline{A}(v) = \overline{A(\overline{v})} = \overline{A(v)} = A(v)$, which means $A(v) \in V$. If we write

$$A = \frac{A+\overline{A}}{2} + i\frac{A-\overline{A}}{2i} =: A_1 + iA_2$$

a quick conclusion is that $\overline{A_1} = A_1, \overline{A_2} = A_2$, which tells us that both $A_1, A_2 \in \mathbf{Hom}(V,W)$ when restricted to V.

Intuitively, we may guess that there is an identification between the complexification $\mathbf{Hom}(V,W)_{\mathbb{C}}$ and $\mathbf{Hom}(V_{\mathbb{C}}, W_{\mathbb{C}})$ s.t. the conjugation of the complexification space coincide with the conjugation of the linear mapping .For arbitrary $A_1, A_2 \in \mathbf{Hom}(V,W)$ and $v_1, v_2 \in V$, the identification is given by

$$(A_1 + iA_2)(v_1 + iv_2) := (A_1v_1 - A_2v_2) + i(A_1v_2 + A_2v_1)$$

For each $v \in V$, a short computation shows that

$$\begin{aligned}\overline{A_1 + iA_2}(v) &= \mathrm{Con} \circ (A_1 + iA_2) \circ \mathrm{Con}(v) \\ &= \mathrm{Con} \circ (A_1 + iA_2)(v) \\ &= \mathrm{Con}(A_1v + iA_2v) \\ &= A_1v - iA_2v \\ &= (A_1 - iA_2)v, \\ \overline{A_1 + iA_2}(iv) &= \mathrm{Con} \circ (A_1 + iA_2) \circ \mathrm{Con}(iv) \\ &= \mathrm{Con} \circ (A_1 + iA_2)(-iv) \\ &= \mathrm{Con}(A_2v - iA_1v) \\ &= A_2v + iA_1v \\ &= (A_1 - iA_2)(iv)\end{aligned}$$

which by linearity implies that $\overline{A_1 + iA_2} = A_1 - iA_2$, i.e. such identification preserves the conjugation; Conversely, we have deduce a decompositon for each linear mapping as above, this shows that the identification is indeed an isomorphism.

2.3 Complexification of real Lie algebra

Real Lie algebra is a special kind of real vector space equipped with a special bilinear form $[\cdot,\cdot]$, we first verify that the extension of the real Lie bracket is a complex Lie bracket.

Proposition 2.7. *The unique extension* $[\cdot,\cdot]_{\mathbb{C}}$ *of* $[\cdot,\cdot]$ *makes the complexification* $\mathfrak{g}_{\mathbb{C}}$ *of* $\mathfrak{g}$ *a complex Lie algebra.*

Proof. It remains to show that Jacobi Identity holds for $[\cdot,\cdot]$, that is,

$$[X,[Y,Z]_{\mathbb{C}}]_{\mathbb{C}} = [[X,Y]_{\mathbb{C}},Z]_{\mathbb{C}} + [Y,[X,Z]_{\mathbb{C}}]_{\mathbb{C}}$$

First consider the case that $X = X_1 + iX_2 \in \mathfrak{g}_{\mathbb{C}}$ and $Y, Z \in \mathfrak{g}$, then by linearity we can deduce

$$\begin{aligned}
[X_1 + iX_2, [Y,Z]_{\mathbb{C}}]_{\mathbb{C}} &= [X_1,[Y,Z]] + i[X_2,[Y,Z]] \\
&= ([[X_1,Y],Z] + [Y,[X_1,Z]]) \\
&+ i([[X_2,Y],Z] + [Y,[X_2,Z]]) \\
&= [[X_1 + iX_2, Y]_{\mathbb{C}}, Z]_{\mathbb{C}} + [Y,[X_1 + iX_2, Z]_{\mathbb{C}}]_{\mathbb{C}}
\end{aligned}$$

Furthermore, let $Y = Y_1 + iY_2 \in \mathfrak{g}_{\mathbb{C}}$, the same arguement still holds, which also goes for the case that $Z = Z_1 + iZ_2 \in \mathfrak{g}_{\mathbb{C}}$, this completes the proof. $\square$

Now let's turn our attention to martix real Lie algebra $\mathfrak{g} \subset M(n,\mathbb{C})$, which has a concrete complexification $X + iY$ for any $X, Y \in \mathfrak{g}$. However, such "complexification" doesn't necessarily coincide with the abstract one . In fact, such identification holds with certain extra condition.

Proposition 2.8. *Suppose $\mathfrak{g}$ be as above and that for any $X \in \mathfrak{g}$, $iX \in \mathfrak{g} \Leftrightarrow X = 0$, where iX is the uaual scalar multiplication, then there exists a Lie algebra isomorphism between $\mathfrak{g}_{\mathbb{C}}$ and the matrixs $\{X + iY | X, Y \in \mathfrak{g}\}$.*

Proof. In order to make no confusion, we denote (X,Y) as the abstract complexification and check that $\varphi : (X,Y) \mapsto X + iY$ is right the desired isomorphism. It's not hard to observe that φ is a Lie algebra homomorphism by computation. If $\varphi(X_1,Y_1) = \varphi(X_2,Y_2)$, then we can deduce that $X_1 - X_2 = i(Y_2 - Y_1) \in \mathfrak{g}$, which, by hypothesis, implies that $X_1 - X_2 = Y_1 - Y_2 = 0$, i.e. φ is injective, thus a isomorphism. $\square$

Using the proposition above and the arguement from Introduction, we can know that $\mathfrak{su}(2)$ has a concrete complexification, with basis being

$$\begin{aligned}
(X,Y,H) &:= (\sigma_1,\sigma_2,\sigma_3)\begin{pmatrix} 1 & -1 & 0 \\ -i & -i & 0 \\ 0 & 0 & -2i \end{pmatrix} \\
&= (\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix})
\end{aligned}$$

which is right the basis of $\mathfrak{sl}(2,\mathbb{C})$. What's more, the injectivity gives us a uniquee way to decompose arbitrary $A \in \mathfrak{sl}(2,\mathbb{C})$, that is

$$A = \frac{A - A^\dagger}{2} + i\frac{A + A^\dagger}{2i}$$

where $A^\dagger$ is the conjugate transpose of A.

3 Irreducible Representation Of $\mathfrak{su}(2)$

3.1 Reducing the problem by complexification

By theorem 2.3, we can obtain a diagram for a real representation f for a real Lie algebra $\mathfrak{g}$ onto a real vector space W:

$$\begin{tikzcd}
\mathfrak{g} \arrow[r, "f"] \arrow[d, "\iota"'] & \mathfrak{gl}(W) \arrow[d, "\iota"'] \\
\mathfrak{g}_{\mathbb{C}} \arrow[r, "\tilde{f}"] & \mathfrak{gl}(W_{\mathbb{C}}) \cong \mathfrak{gl}(W)_{\mathbb{C}}
\end{tikzcd}$$

If f is irreducible, we can obtain that

$$\begin{aligned}
\tilde{f}(\mathfrak{g}_{\mathbb{C}}).W_{\mathbb{C}} &= \tilde{f}(\mathbb{C}.\mathfrak{g}).(\mathbb{C}.W) \\
&= \mathbb{C}.(f(\mathfrak{g}).W) \\
&= \mathbb{C}.W = W_{\mathbb{C}}
\end{aligned}$$

which shows that the induced complex representation of the complexification is also irreducible. Therefore, we just need to study the irreducible representation for the complexification and then restrict it to the 'real' subspace.

3.2 Irreducible representation of $\mathfrak{sl}(2, \mathbb{C})$

We first recall some conclusion for semisimple Lie algebra. If $\mathfrak{g} \subset M(n, \mathbb{C})$ is semisimple, then its abstract Jordan-Chevalley decompositon coincide with the regular one; On the other hand, all the homomorphism image of a semisimple Lie algebra is semisimple, and the decompositon of each image is given by the image of the semisimple and nilpotent component of the preimage. Hence we know that for any m-dimensional vector space and the standard basis $\{X, Y, H\}$ for $\mathfrak{sl}(2, \mathbb{C})$, X, Y act nilpotently while H is diagnoalizable when acting on V. By the commutation relation

$$\begin{aligned}
[H, X] &= 2X \\
[H, Y] &= -2Y \\
[X, Y] &= H
\end{aligned}$$

we have the theorem for $\mathfrak{sl}(2, \mathbb{C})$.

Theorem 3.1. *For any m-dimensional vector space V, there exists an irreducible representation for $\mathfrak{sl}(2, \mathbb{C})$.*

Proof. We first look at the properties of the irreducible representation. Since H acts diagnoally on V, we can decompose it into several eigensubspace of H. For each eigenvector $u \in V$ with eigenvalue λ, we can obtain that

$$\begin{aligned}
H.X.u = X.H.v + [H, X].v = (\lambda + 2)X.v \\
H.Y.u = Y.H.v + [H, Y].v = (\lambda - 2)Y.v
\end{aligned}$$

This is equivalent to saying that for each eigensubspace V_λ, we have

$$\begin{aligned} X.V_\lambda &\subset V_{\lambda+2} \\ Y.V_\lambda &\subset V_{\lambda-2} \end{aligned}$$

This result drive us to find out a finite chain structure related to the integer 2. We define an equivalent relation for H's eigenvalue by

$$\mu \sim \nu \Longleftrightarrow \mu - \nu \in 2\mathbb{Z}$$

For each equivalent class, since it's finite, we can always find the unique element $\lambda_{\max}$ such that for any other element λ in the same class, $\lambda_{\max} - \lambda$ is always positive. Now we devide the eigenvalues of H into some disjoint union with such maximal element, namely

$$V = \bigoplus_{i=1}^{s} \bigoplus_{k=0}^{m_i} V_{\lambda_i - 2k}$$

Now choose a nonzero $u_i^0 \in V_{\lambda_i}$, and let $u_i^k := Y^k u_i^0$, by the previous arguement we know that $Xu_i^0 = 0$ and $u_i^{m_i+k} = Y^{m_i+k}u_i^0 = 0$ for any natural number k. Apply the equation $[X, Y] = XY - YX = H$ to u_i^k, we obtain a recursion relation for Xu_i^k:

$$Xu_i^{k+1} = YXu_i^k + (\lambda_i - 2k)u_i^k$$

Let k exhaust from 0 to m_i, we have a series of equation:

$$\begin{aligned} &Xu_i^{m_i+1} = YXu_i^{m_i} + (\lambda_i - 2m_i)u_i^{m_i} \\ &Xu_i^{m_i} = YXu_i^{m_i-1} + (\lambda_i - 2(m_i - 1))u_i^{m_i-1} \\ &\dots \\ &Xu_i^1 = YXu_i^0 + \lambda_i u_i^0 \end{aligned}$$

Apply Y^{k-1} to the kth equation, and sum them up, we can obtain

$$\begin{aligned} 0 &= Xu_i^{m_i+1} \\ &= Y^{m_i+1}Xu_i^0 + \sum_{k=0}^{m_i} (\lambda_i - 2(m_i - k))u_i^{m_i} \\ &= (m_i + 1)(\lambda_i - m_i) \end{aligned}$$

which implies that $\lambda_i = m_i$. In fact, the computation above shows that for $0 \leq k \leq m_i$, we have

$$Xu_i^{k+1} = (k + 1)(m_i - k)u_i^k$$

that is, $u_i^k \neq 0$. This gives us a group of linearly independent vectors since they sit in different eigensubspace. Then the subspace spanned by $\{u_i^k\}_{k=0}^{m_i}$ is invariant, which forces it to be the whole space V by irreducibility, and $\lambda_i = m_i = \dim V - 1$. Thus we have a explicit construction of irreducible $\mathfrak{sl}(2, \mathbb{C})$ representation on arbitrary $m+1$-dimensional vector space V, for convience we replace H by $\frac{H}{2}$ as another basis, the matrix under some basis $\{u_0, Y.u_0, \dots, Y^m.u_0\}$ is given by

$$H \mapsto \begin{pmatrix} \frac{m}{2} & 0 & 0 & \cdots & 0 \\ 0 & \frac{m}{2}-1 & 0 & \cdots & 0 \\ 0 & 0 & \frac{m}{2}-2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & -\frac{m}{2} \end{pmatrix}$$

$$X \mapsto \begin{pmatrix} 0 & m & 0 & \cdots & 0 \\ 0 & 0 & 2(m-1) & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}$$

$$Y \mapsto \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{pmatrix}$$

□

Consider a special cases that $m = 2$, then we obtain a 3-dimensional representation for $\mathfrak{su}(2)$ under the basis $\{\sqrt{2}u_0, Y.u_0, \frac{Y^2.u_0}{\sqrt{2}}\}$:

$$\sigma_1 \mapsto \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & 0 \end{pmatrix}$$

$$\sigma_2 \mapsto \begin{pmatrix} 0 & \frac{-i}{\sqrt{2}} & 0 \\ \frac{i}{\sqrt{2}} & 0 & \frac{-i}{\sqrt{2}} \\ 0 & \frac{i}{\sqrt{2}} & 0 \end{pmatrix}$$

$$\sigma_3 \mapsto \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$